**25901051** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-BVOC (HONOURS) Regular EXAMINATION November 2025

## First SEMESTER

## Skill Development Components (SDC) - MG1SDCREM100 - ELECTRICAL FUNDAMENTALS AND ELECTRONICS

(2025 ADMISSION ONWARDS)

**Duration: 2 Hours** **Maximum Marks: 70**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

Short answer

Answer any **25** questions

Each question carries **1** marks

1. Show the dielectric material in capacitors. [K] / [CO1]
2. Recall the unit of resistance. [K] / [CO1]
3. Define a relay. [K] / [CO1]
4. Recognize the effect of increasing inductance on current. [K] / [CO1]
5. Identify another name for variable resistor. [K] / [CO1]
6. Identify the construction parts of a coaxial cable. [K] / [CO1]
7. List two examples of voltage sources. [K] / [CO1]
8. Identify the unit of inductance. [U] / [CO2]
9. Classify the components of an LCR circuit. [A] / [CO3]
10. Explain susceptance. [U] / [CO2]

11. Write the full form of LED. [A] / [CO4]

12. Write the formula for power in AC circuits [A] / [CO3]

13. Differentiate between AC and DC. [U] / [CO2]

14. Write the equation of a sinusoidal AC voltage [A] / [CO3]

15. Discuss the main purpose of Norton's theorem. [U] / [CO2]

16. Write the power in a three-phase system. [A] / [CO3]

17. Present one example of a semiconductor material. [A] / [CO5]

18. Indicate one electrical property of metals. [U] / [CO4]

19. Write the three terminals of a transistor. [A] / [CO5]

20. Illustrate forward biasing in a P-N junction. [A] / [CO4]

21. Show what an amplifier increases in a signal. [A] / [CO5]

22. Explain the type of current allowed by a diode [U] / [CO4]

23. Write one example of an electrical insulator. [A] / [CO5]

24. Describe conduction electrons. [U] / [CO4]

25. Write the concept of a half-wave rectifier. [A] / [CO5]

26. Explain the concept of doping in semiconductors. [U] / [CO4]

27. Write the meaning of mass number. [A] / [CO5]

**[1x25 = 25]**

## Part B

Short Essay

Answer any **5** questions

Each question carries **5** marks

28. State the consruction of reley. [K] / [CO1]

29. List the different types of switches and explain toggle switch. [K] / [CO1]

30. Explain the application of LCR circuits in resonance experiments and power factor improvement. [U] / [CO2]

31. Explain the working of a Zener diode as a voltage regulator. [U] / [CO4]

32. Demonstrate how to find the Thevenin equivalent of a simple network. [U] / [CO2]

33. Describe the structure and behavior of electrons in an atom. [U] / [CO4]

34. Classify the different types of phototransistors and present their working differences based on sensitivity and response time. [A] / [CO5]

**[5x5 = 25]**

## Part C

Essay

Answer any **2** questions

Each question carries **10** marks

35. Define a rocker switch and state its construction and working principle with a neat diagram [K] / [CO1]

36. Write short notes on practical applications of series and parallel LCR circuits, including Q-factor and power factor. [A] / [CO3]

37. Model a simple circuit and show how the maximum power transfer theorem is applied to find maximum power delivered to a load. [A] / [CO3]

38. Explain what valence electrons are and describe how they help in forming bonds between atoms. [U] / [CO4]

**[10x2 = 20]**